

An Exact Five-Dimensional Counterexample to Concavity of Determinantal-Process Entropy

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Abstract

For a determinantal probability measure on a finite labelled set, Russell Lyons conjectured that the Shannon entropy of the full subset-valued law is concave in the positive-contraction kernel. We give an exact counterexample on five coordinates over a complex Hilbert space. The midpoint is a rational real symmetric kernel, the chord has a rational purely imaginary Hermitian direction, and every claim is reduced to finite rational arithmetic. Sylvester certificates put the midpoint and both endpoints strictly between zero and the identity. A table gives all 32 atom probabilities at the midpoint and endpoints; every entry is positive, every distribution sums to one, and the two endpoint distributions agree entry by entry. Directed rational enclosures for logarithms prove

$$H(K + hA) - H(K) > 10^{-13} \quad \text{and} \quad \frac{1}{200} < D^2 H(K)[A, A] < \frac{1}{190}.$$

Thus the complex-Hilbert-space version of the conjecture is false. The real-symmetric restriction and the minimum possible counterexample dimension remain open.

Draft status. This is a private mathematical note awaiting an independent paper-level review. All load-bearing prose and proof organization introduced here are marked `NEW_REWRITE_REQUIRES_VERIFICATION` in `source_map.md`.

1 The conjecture and the exact conclusion

Let $E = [n]$ be a labelled set and let K be a Hermitian operator on $\ell^2(E; \mathbb{C})$ with $0 \preceq K \preceq I$. The determinantal probability measure $\mathbb{P}^K$ is characterized by

$$\mathbb{P}^K(T \subseteq X) = \det K[T] \quad (T \subseteq E), \tag{1}$$

where $K[T]$ is the principal submatrix on the fixed coordinates T . Write $p_K(S) = \mathbb{P}^K(X = S)$ for its 2^n atom probabilities and

$$H(K) = - \sum_{S \subseteq E} p_K(S) \log p_K(S), \quad 0 \log 0 := 0. \tag{2}$$

Conjecture 2.6 of Lyons [1] asserts midpoint concavity of this quantity on the convex set of positive contractions. The source explicitly allows real or complex Hilbert spaces, defines a positive contraction to be self-adjoint with $0 \preceq K \preceq I$, and uses the entropy of the complete subset-valued distribution. These points occur, respectively, on printed pages 138, 140, and 141; the coordinate DPP definition begins on printed page 137.

Theorem 1. *On $E = [5]$, there exist strict positive contractions $K - hA$ and $K + hA$ on $\ell^2(E; \mathbb{C})$, with midpoint K , such that*

$$H(K) < \frac{H(K - hA) + H(K + hA)}{2}.$$

More precisely, the matrices in (3)–(4) with $h = 1/100000$ satisfy

$$H(K + hA) - H(K) > 10^{-13}, \quad \frac{1}{200} < D^2 H(K)[A, A] < \frac{1}{190}.$$

Consequently the full complex-Hermitian version of Lyons's entropy-concavity conjecture is false.

The assertion concerns the labelled-coordinate entropy in (2). No invariance under an arbitrary unitary change of basis is used or asserted.

2 The rational chord

Define

$$K = \frac{1}{10000} \begin{pmatrix} 2719 & -3449 & 1009 & -818 & -2490 \\ -3449 & 5840 & -2889 & -1513 & 1322 \\ 1009 & -2889 & 2148 & 2505 & 1101 \\ -818 & -1513 & 2505 & 4698 & 3958 \\ -2490 & 1322 & 1101 & 3958 & 4596 \end{pmatrix} \quad (3)$$

and $A = iB$, where

$$B = \frac{1}{50} \begin{pmatrix} 0 & -3 & -13 & -16 & 4 \\ 3 & 0 & -4 & 15 & 17 \\ 13 & 4 & 0 & 6 & -15 \\ 16 & -15 & -6 & 0 & 1 \\ -4 & -17 & 15 & -1 & 0 \end{pmatrix}, \quad h = \frac{1}{100000}. \quad (4)$$

Since $B^T = -B$, the matrix $A = iB$ is Hermitian. The Pfaffian numerator of the leading 4×4 block of $50B$ is $(-3)(6) - (-13)(15) + (-16)(-4) = 241$, so that block has determinant $58081/6250000 > 0$. An odd-dimensional real antisymmetric matrix has zero determinant; hence $\text{rank } A = \text{rank } B = 4$.

3 Strict feasibility by Sylvester's criterion

Lemma 1. *The three kernels K , $K - hA$, and $K + hA$ satisfy $0 < K_t < I$ strictly.*

Proof. For each row below, the five entries are the leading principal minors in orders

$1, \dots, 5$:

$$\begin{aligned}
K &: \frac{2719}{10^4}, \frac{3983359}{10^8}, \frac{24692191}{10^{12}}, \\
&\quad \frac{3908717}{5 \cdot 10^{14}}, \frac{118205061}{10^{20}}; \\
I - K &: \frac{7281}{10^4}, \frac{18393359}{10^8}, \frac{59312197809}{10^{12}}, \\
&\quad \frac{35054548217}{5 \cdot 10^{14}}, \frac{1709337154939}{10^{20}}; \\
K + hA &: \frac{2719}{10^4}, \frac{995839749991}{25 \cdot 10^{12}}, \frac{123460921839}{5 \cdot 10^{15}}, \\
&\quad \frac{4883537625128308081}{625 \cdot 10^{24}}, \frac{7353066474454085031}{625 \cdot 10^{28}}; \\
I - K - hA &: \frac{7281}{10^4}, \frac{4598339749991}{25 \cdot 10^{12}}, \frac{296560989039361}{5 \cdot 10^{15}}, \\
&\quad \frac{43818177021217628308081}{625 \cdot 10^{24}}, \frac{106676909868459401834969}{625 \cdot 10^{28}}.
\end{aligned}$$

Every displayed number is a positive rational number, and the imaginary part of each determinant is exactly zero. Sylvester's criterion proves $K > 0$, $I - K > 0$, $K + hA > 0$, and $I - K - hA > 0$. Complex conjugation gives $K - hA = \overline{K + hA}$ and $I - K + hA = \overline{I - K - hA}$, so their corresponding leading principal minors are the same. Thus both remaining Hermitian matrices are positive definite as well. $\square$

4 All atom probabilities

For $S \subseteq [5]$, put $C = S^c$, let I_C denote the diagonal coordinate projection onto C , and set

$$M_S = K - I_C, \quad p_S(t) = (-1)^{|C|} \det(M_S + tiB). \quad (5)$$

Lemma 2. *For every real t for which $K + tA$ is a positive contraction, $p_S(t) = \mathbb{P}^{K+tA}(X = S)$.*

Proof. Möbius inversion of (1) gives

$$\mathbb{P}^{K+tA}(X = S) = \sum_{T \supseteq S} (-1)^{|T \setminus S|} \det(K + tA)[T]. \quad (6)$$

Expanding $\det(K + tA - I_C)$ by the diagonal entries contributed by $-I_C$ leaves exactly the same principal minors, with the same signs after multiplication by $(-1)^{|C|}$. This proves (5). $\square$

Table 1 lists the exact data for all 32 masks; a bit equal to one means that the corresponding coordinate belongs to S . All center numerators c_S and endpoint numerators e_S are strictly positive, and direct integer summation gives

$$\sum_S c_S = 10^{20}, \quad \sum_S e_S = 625 \cdot 10^{28}. \quad (7)$$

Thus the center and endpoint rows are positive probability distributions and each sums exactly to one. Moreover,

$$p_S(-h) = \overline{p_S(h)} = p_S(h),$$

because the matrices in (5) are Hermitian and changing h to $-h$ complex-conjugates them. Hence the two endpoint distributions agree for each labelled subset, not merely after a permutation or in aggregate.

5 The exact logarithm certificates

We now describe the directed interval calculation used for both strict inequalities. It involves rational arithmetic only. For a positive rational x , write uniquely $x = 2^k y$ with $1 \leq y < 2$, and put $z = (y - 1)/(y + 1) \in [0, 1/3]$. For $N = 80$, define

$$L_N(y) = 2 \sum_{m=0}^{N-1} \frac{z^{2m+1}}{2m+1}, \quad R_N(y) = \frac{2z^{2N+1}}{(2N+1)(1-z^2)}.$$

The positive-term expansion of $\operatorname{atanh} z$ gives the rational enclosure

$$L_N(y) \leq \log y \leq L_N(y) + R_N(y). \quad (8)$$

The same formula at $y = 2$, $z = 1/3$, encloses $\log 2$. Combining the intervals for $\log y$ and $k \log 2$ uses the lower endpoint of the $\log 2$ interval when $k \geq 0$ and the upper endpoint when $k < 0$; the opposite endpoints give the upper bound. Finally, in a sum $\sum_j a_j \log x_j$, the lower bound uses the lower logarithm endpoint when $a_j > 0$ and the upper endpoint when $a_j < 0$. The upper bound uses the reverse choices. This explicit sign rule is what makes the certificates directed.

5.1 The midpoint Hessian

Since M_S is real symmetric and B is real antisymmetric, $\operatorname{tr}(M_S^{-1}B) = 0$. The determinant derivative formula applied to (5) therefore gives

$$p'_S(0) = 0, \quad p''_S(0) = p_S(0) \operatorname{tr}(M_S^{-1} B M_S^{-1} B). \quad (9)$$

The last column of Table 1 lists the numerator d_S of every $p''_S(0)$ over the common denominator 312500000000000. Exact summation gives $\sum_S p''_S(0) = 0$. Twice differentiating (2), using $p'_S(0) = 0$, yields

$$D^2 H(K)[A, A] = - \sum_S p''_S(0) \log p_S(0) = \sum_S q_S \log p_S(0), \quad q_S = -p''_S(0). \quad (10)$$

Applying the directed rule above to the 32 rational logarithms gives the strict rational enclosure

$$\frac{5205948204140858303}{10^{21}} < D^2 H(K)[A, A] < \frac{5205948204140858304}{10^{21}}. \quad (11)$$

The lower endpoint exceeds $1/200$ by $205948204140858303/10^{21}$, while $1/190$ exceeds the upper endpoint by $16984126895682691/29687500000000000000$. Both margins are positive, which proves the Hessian claim without using its decimal approximation.

5.2 The finite chord

Use the 64 rational logarithmic terms in

$$H(K + hA) - H(K) = - \sum_S p_S(h) \log p_S(h) + \sum_S p_S(0) \log p_S(0). \quad (12)$$

The probabilities are exactly those in Table 1. Applying (8), with each coefficient directed according to its sign, gives

$$\frac{233555253804221762}{10^{30}} < H(K + hA) - H(K) < \frac{233555253804221763}{10^{30}}. \quad (13)$$

The lower endpoint exceeds 10^{-13} by

$$\frac{66777626902110881}{500000000000000000000000000000} > 0.$$

Because the two endpoint distributions agree term by term, $H(K - hA) = H(K + hA)$. Lemma 1 and (13) now prove Theorem 1.

6 Scope and compatibility

The counterexample uses a real symmetric midpoint but a purely imaginary rank-four Hermitian direction. It therefore settles the complex-Hilbert-space branch of the original conjecture, whose quantifiers include this chord. It does not settle a separately imposed restriction to real symmetric kernels. It also does not show that five is the least possible counterexample dimension.

This result is compatible with the independently verified fixed-complement-block star-fibre theorem. In that theorem the kernels have the form

$$\begin{pmatrix} a & v^* \\ v & C \end{pmatrix}$$

with the same Hermitian block C along the fibre, so a difference within one fibre is supported on one coordinate star and has rank at most two. The present direction has rank four and is not a difference inside such a fibre. Coordinate dependence also prevents an arbitrary unitary change of basis from turning this observation into a claim about the same entropy functional.

The exact certificate proves a counterexample, but it does not establish absolute priority or novelty. The source audit found no earlier public proof or counterexample in its bounded search as of 5 September 2026; that limited result is not a systematic or exhaustive priority determination.

Data availability

The directory `certificates/` contains the two frozen 32-row rational tables and their JSON summaries. Table 1 is a typeset projection of those fixed tables, not a floating-point recomputation.

A The 32 exact atoms and second derivatives

Table 1: Exact event data. For each mask S , the center probability is $c_S/10^{20}$, either endpoint probability is $e_S/(625 \cdot 10^{28})$, and $p''_S(0) = d_S/31250000000000$. Thus $q_S = -d_S/31250000000000$ in the Hessian formula.

mask	c_S	e_S	d_S
00000	1709337154939	106676909868459401834969	-156662315231953
00001	2656283230935061	166017744649677366641455031	42716236056953
00010	6115970682105061	382248259084754513622575031	91453188204453
00011	3971055587349804939	248190974216992099660334134969	7629290970547
00100	3263125734325061	203945390776741244180575031	32381424934453
00101	4816014638897584939	301000914895668312929776134969	-35430745759453
00110	4184669109046414939	261541819326161410782795014969	10760477093047
00111	246757521675061	154223489501735043248275031	6902443731953
01000	8434437215975061	5271523984601116887232165031	72461670379453

mask	c_S	e_S	d_S
01001	12094797748615934939	755924859218562692485233544969	-69933241204453
01010	25132478169064764939	1570779885473979040338252424969	-92568768351953
01011	4467094503325061	279193455111755487790865031	48653939176953
01100	3805671766912544939	237854485447237453607694424969	15203394918047
01101	2116290055545061	132268155485142218348865031	27013575906953
01110	5789712006715061	361857023390619365329985031	22970928054453
01111	781625194939	48828023184808626724969	-23551498879453
10000	7009200306245061	438075093302307823678975031	74161991514453
10001	6285970421125664939	392873151332436946350277734969	12082887660547
10010	25078103510574494939	1567381469304727819203296614969	-106178114486953
10011	5366317393595061	335394862896526622746675031	25796835311953
10100	8650505306722274939	540656581615572207472738614969	-54569976216953
10101	1730314645815061	108144694433938353304675031	29070497041953
10110	5070067596985061	316879269464890500285795031	44653324189453
10111	861634924939	53827415363673670914969	-24767445014453
11000	5915774434040624939	369735902130528962028196024969	2989903338047
11001	3354766127465061	209672901149433797847265031	18182867486953
11010	4413568778635061	275848120757135944828385031	72092444634453
11011	781653274939	48810450868229128324969	-42878815459453
11100	2699019730855061	168688744962347675386385031	11783906364453
11101	537301054939	33559863856498570324969	-21452077189453
11110	892249884939	55724684329351589204969	-40933479336953
11111	118205061	7353066474454085031	-34749838047

References

- [1] Russell Lyons. Determinantal probability: Basic properties and conjectures, 2014. Proceedings of the International Congress of Mathematicians 2014, volume IV, pages 137–161.